Course code : PHY-1101
Course Title :Physics I (Mechanics, Waves, Optics, Heat and Thermodynamics)
Credit Hours: 3 **Contact Hours:** 3 per Week

Objectives:
This course is designed to provide students in Applied Science who wish to study engineering at university with an enhanced background in order to improve their chances of success in their chosen program. The material will be presented using the normal mix of lectures and laboratory experiments and demonstrations. This course objective is also to

- Study some of the applications of physics to technology and other scientific fields.
- Give a solid grounding in basic physics that will serve as a basis for further study in Engineering.
- Develop analytical and mathematical skills that will be applicable to many scientific endeavours.
- Demonstrate the application of the scientific method through laboratory experiments.
- Develop an appreciation for how scientific measurements are made.
- Assess the role of physics, in helping us to better understand the complex, technological society.

Learning Outcome:
Course outcomes are statements that describe what students are expected to know and be able to do by the time of graduation. These outcomes relate to the skills, knowledge, and behaviors that students acquire in their matriculation through the program. The ultimate goal of the course outcomes is to encourage achievement of the course educational objectives.

Upon completion of a Course of Study, students should be able to demonstrate:

- an ability to apply knowledge of mathematics, science, and engineering.
- an ability to design and conduct experiments, as well as to analyze and interpret data.
- an ability to identify, formulate, and solve physics and engineering problems.
- the broad education necessary to understand the impact of physics and engineering solutions in a global, economic, environmental, and societal context.
- an ability to use the techniques, skills, and modern tools necessary for physics and engineering careers.
- an ability to analyze the engineering concepts based on fundamental physical concepts.

Section –A
(Mid-term Exam: 30 Marks)

Mechanics :

1. **Dynamics of Rigid Body:** Linear motion of a body as function of time, position and velocity, momentum, conservation theorem of momentum and energy, collision and torque, center of mass of rigid body, rotational kinetic energy, fly wheel, axes theorems and their applications.
2. **Gravity and Gravitation:** Definitions, compound pendulum, gravitational potentials and fields and relation between them, potential due to spherical shell, escape velocity and Kepler's law of planetary motion.
3. **Introductory quantum mechanics:** Wave function, Uncertainty principle, Postulates, Schrodinger time independent equation, Expectation value, Probability, Particle in a zero potential, Calculation of energy.

Section- B (Final Exam: 50 Marks)
Group- A (20-Marks)
Waves & Oscillations:

Waves: Waves in elastic media, Differential equation of a progressive wave, Power and intensity of wave motion, standing waves , Sound waves, beats and Doppler's effect in sound, Group velocity and phase velocity.

Oscillation: Differential equation of a simple harmonic oscillator, Total energy and average energy, Combination of simple harmonic oscillations, Lissajous figures, Spring mass system, Calculation of time period of torsional pendulum, Damped oscillation, Forced oscillation, Resonance.

Group-B (30 Marks)

Heat and Thermodynamics: Thermodynamic system, first and second law of thermodynamics and their applications, the thermodynamic temperature scale, Carnot's heat engine, the efficiency of engine, combined first and second law of thermodynamics, entrop, Entropy, Thermodynamics functions, Maxwell relations, Third Law of thermodynamics.

Optics:

Interference: Defects of images: spherical aberration, astigmatism, coma, distortion, curvature, chromatic aberration. Theories of light; Interference of light: Young‟s double slit experiment, interference in thin films, Newton's rings, interferometers;

Diffraction & Polarization: Diffraction by single slit, diffraction from a circular aperture, resolving power of optical instruments, diffraction at double slit and N-slits, diffraction grating; polarization: Production and analysis of polarized light, Brewster‟s law, Malus law, polarization by double refraction.

Reccommended Books

1. Dr. Giasuddin Ahmad : Engineering Physics (Part-1)
2. David Halliday and Robert Resnick : Physics part-I.
3. Young and Fredman : University Physics,
4. D.S. Mathur : Properties of matter
5. Brij lal and N. Subrahmanyum : A textbook of sound
6. Brij lal and N. Subrahmanyum : Heat and thermodynamics
7. Brij lal and N. Subrahmanyum : Properties of matter
8. F.W.Sears : Thermodynamics
9. F.W.Constan : Theoretical Physics
10. Modern Physics : Arthur Beiser
11. Quantum Mechanics : Gupta Kumar Sharma